

Article

Torque Ripple Suppression in BLDC Reaction Wheels Using Adaptive Composite Control Strategy Under Non-Ideal Back-EMF

Zhicheng Wang ¹, Haitao Li ¹, Tong Wen ^{1,*}, Haitao Li ¹ and Xiangwen Chen ^{2,*}

¹ School of Instrumentation and Optoelectronic Engineering, Beihang University, Beijing 100080, China; zy2354213@buaa.edu.cn (Z.W.); haitaoli@buaa.edu.cn (H.L.); lht20030718@buaa.edu.cn (H.L.)

² Ningbo Innovation Institute, Beihang University, Ningbo 315000, China

* Correspondence: wentong@buaa.edu.cn (T.W.); xiangwen@buaa.edu.cn (X.C.)

Abstract

High-precision torque regulation is essential to ensure reaction wheel systems meet the stringent attitude control requirements of modern spacecraft. In three-phase half-bridge brushless DC (BLDC) drives, non-ideal back-electromotive force (back-EMF) waveforms cause pronounced conduction interval torque ripple, leading to inaccurate and unstable output torque. To address this problem, this article proposes a composite torque control strategy integrating an Adaptive Nonsingular Fast Terminal Sliding-Mode Observer (ANFTSMO) with an Adaptive Sliding-Mode Controller (ASMC). The ANFTSMO achieves precise back-EMF estimation and electromagnetic torque reconstruction by eliminating singularities, reducing chattering, and adaptively adjusting observer gains. Meanwhile, the ASMC employs an adaptive switching gain function to achieve asymptotic current convergence with suppressed chattering, thereby ensuring accurate current tracking. System stability is verified via Lyapunov analysis. Simulation and experimental results demonstrate that, compared with conventional constant-current control, the torque smoothness and disturbance rejection of the proposed method are improved, enabling precise and stable reaction wheel torque delivery for high-accuracy spacecraft attitude regulation.

Keywords: reaction wheel; BLDC motor; torque ripple; ANFTSMO; ASMC

1. Introduction

With the rapid advancement of space technology, modern space missions require increasingly precise and responsive attitude control systems for spacecraft [1]. Reaction wheels are widely used because they provide continuously adjustable torque with rapid dynamics and no propellant, enabling high-precision regulation [2]. Most reaction wheel drives employ brushless DC (BLDC) motors to ensure their high efficiency, long life, compact size, and digitally amenable construction [3], typically with three-phase half-bridge inverters for hardware simplicity and cost effectiveness relative to full-bridge schemes [4]. However, the half-bridge topology is susceptible to non-ideal back-electromotive force (back-EMF) waveforms [5]. Such quasi-sinusoidal distortions with flattened crests induce pronounced conduction interval torque ripple, degrading torque smoothness, reducing output accuracy, and ultimately impairing the precision of spacecraft attitude control [6]. To mitigate torque ripple in BLDC drives with distorted back-EMF, various approaches have been proposed. Some researchers have focused on structural improvements; for example, Farshid Mahmouditabar et al. [7] optimized the motor design to reduce torque



Academic Editor: Dong Jiang

Received: 24 November 2025

Revised: 22 December 2025

Accepted: 29 December 2025

Published: 3 January 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article

distributed under the terms and

conditions of the [Creative Commons](#)

[Attribution \(CC BY\)](#) license.

ripple. However, in this study, we focus on addressing this issue through control algorithms. Among the available control techniques, harmonic injection methods have been widely explored. For instance, Zhang et al. [8] proposed selective harmonic current injection to cancel dominant harmonics, and He et al. [9] combined harmonic injection with active disturbance rejection control to improve torque smoothness. These methods are effective with full-bridge topologies, where multi-phase conduction allows superimposing auxiliary harmonics. However, in half-bridge BLDC drives with single-phase conduction, such flexibility is absent. Attempting to shape harmonics within the conducting phase increases DC-bus utilization and electromagnetic interference (EMI), making these methods unsuitable for reaction wheel applications. Another method for reducing torque ripple is direct torque control (DTC). Chen et al. [10] incorporated space-vector modulation into the DTC framework to suppress torque and flux oscillations, while Sa et al. [11] refined zero-vector selection and phase-current integration to produce smoother torque profiles. These approaches are effective in sinusoidal permanent magnet synchronous motor (PMSM) drives with full-bridge inverters, where a complete voltage–vector set and reliable flux models are available. However, in half-bridge BLDC drives, the restricted vector set and distorted back-EMF characteristics limit their applicability, rendering the ripple reduction mechanisms less effective in reaction wheel systems. Additionally, researchers have refined commutation accuracy so that the applied current is phase matched to the back-EMF, thereby mitigating torque ripple. Li et al. [12] proposed an adaptive sliding mode observer for rotor tracking at medium and high speeds, and Wang et al. [13] enhanced low-speed robustness by modifying switching laws. These methods improve commutation timing, but, with distorted EMF and fixed 120° conduction, even perfect commutation does not eliminate structural torque ripple.

Although prior methods can reduce torque ripple in some settings, they are not readily transferable to three-phase half-bridge reaction wheel drives, where single-phase conduction and non-ideal back-EMF are prevalent. Moreover, most schemes treat phase current as the primary feedback variable rather than directly regulating electromagnetic torque, leaving conduction interval ripple insufficiently addressed. Motivated by these limitations, this article proposes a composite torque control strategy that uses torque as both the command and the feedback variable. The framework relies on accurate real-time torque reconstruction, enabled by an Adaptive Nonsingular Fast Terminal Sliding-Mode Observer (ANFTSMO), which eliminates singularities, mitigates chattering, and adapts its injection gain to the estimation error to ensure high-fidelity back-EMF estimation under non-ideal waveforms. The inner current loop is implemented by an adaptive-gain sliding-mode controller (ASMC) to achieve robust asymptotic current tracking with reduced chattering and strong robustness to parameter variations to account for mechanical effects beyond the motor. Together, the ANFTSMO and the ASMC compensation form an integrated nonlinear torque control architecture tailored to half-bridge BLDC reaction wheels with distorted back-EMF. Lyapunov-based analysis and experimental validation confirm precise, low-ripple, and disturbance-resilient torque regulation suitable for high-accuracy spacecraft attitude control.

The remainder of this article is organized as follows: Section 2 introduces the BLDC motor model and the three-phase, three-step commutation topology. Section 3 presents the proposed control strategy. Section 4 provides simulation and experimental validation, and Section 5 concludes this article with future research directions.

2. Modeling the Motor Control System and Disturbance Analysis

2.1. Mathematical Modeling of the BLDC Motor

The reaction-wheel drive system studied here consists of a DC power supply, a three-phase half-bridge inverter, and a BLDC motor equipped with Hall-effect sensors and an encoder. The overall configuration is shown in Figure 1. The controller generates pulse-width modulation (PWM) signals to drive the power electronics, and current and voltage sensors, together with position and velocity feedback, close the control loops.

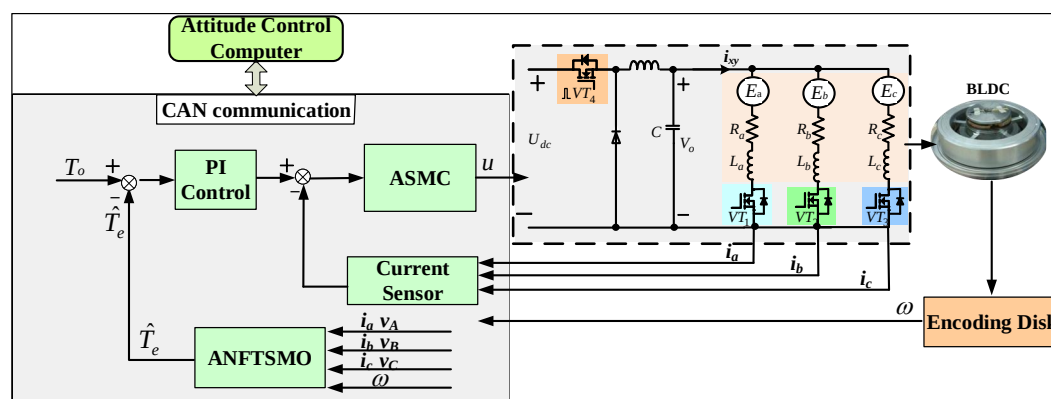


Figure 1. System configuration of the reaction wheel drive.

To facilitate analytical treatment, the BLDC motor is represented as a simplified equivalent three-phase circuit, outlined by the dashed block in Figure 1. In this model, V_0 denotes the BUCK voltage; L_a, L_b, L_c represent the equivalent stator phase inductances; and R_a, R_b, R_c denote the corresponding stator phase resistances. For simplicity, these parameters are assumed to be identical and are expressed as L and R ; The variables E_b and E_c represent the back-EMFs of the A, B, and C phases; VT_1, VT_2, VT_3 , and VT_4 represent switching transistors. The following assumptions are adopted:

- (1) The rotor currents induced by stator harmonic fields are neglected;
- (2) Core and stray losses are ignored;
- (3) The effect of armature reaction is neglected;
- (4) The stator windings are assumed to be identical and symmetrically distributed with a spatial displacement of 120° . Under these assumptions and the equivalent circuit, the phase voltage model of the BLDC motor can be expressed as

$$\begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} E_a \\ E_b \\ E_c \end{bmatrix} \quad (1)$$

where i_a, i_b, i_c represent the corresponding phase currents, and u_a, u_b, u_c are the phase terminal voltages. Based on the principle of power conservation, the mechanical output power P_m is equal to the electromagnetic input power P_e . The electromagnetic power is expressed as the summation of the products of the back-EMFs and the corresponding phase currents [14]:

$$P_m = P_e = e_a i_a + e_b i_b + e_c i_c \quad (2)$$

Neglecting the mechanical losses of the system, including frictional resistance, the mechanical output power of the BLDC motor can be expressed, based on fundamental mechanics, as the product of the electromagnetic torque and the rotor speed, i.e., [15]

$$P_m = T_e \omega_m \quad (3)$$

where T_e denotes the electromagnetic torque, and ω_m represents the mechanical angular velocity.

2.2. Torque Ripple Analysis of Reaction Wheel Systems

For high-precision reaction wheel actuators, torque ripple directly translates into attitude disturbances and jitter. Thus, understanding and mitigating back-EMF distortion are essential for reliable torque generation.

Conventional BLDC commutation regulates constant-phase current with 120° conduction. Let the two-level gating function over one electrical period 2π be defined as

$$S(\theta_e) = \begin{cases} +1, & |\theta_e| \leq \frac{\pi}{3}, \\ 0, & \frac{\pi}{3} < |\theta_e| \leq \pi. \end{cases} \quad (4)$$

where θ_e is the electrical angle.

Under three-phase half-bridge commutation with 120° single-phase conduction, the phase currents are described as

$$i_k(\theta_e) = I S(\theta_e - \zeta_k), \quad I > 0, k \in \{a, b, c\}. \quad (5)$$

where $\zeta_k = \{0, -\frac{2\pi}{3}, \frac{2\pi}{3}\}$ denotes the phase displacement. Consequently, the instantaneous electromagnetic torque is

$$T_e(\theta_e) = \frac{I}{\omega_m} \mathbf{e}^T(\theta_e) \boldsymbol{\sigma}(\theta_e), \quad (6)$$

where $\mathbf{e} = [e_a, e_b, e_c]^T$ and $\boldsymbol{\sigma}(\theta_e) = [S(\theta_e - \zeta_a), S(\theta_e - \zeta_b), S(\theta_e - \zeta_c)]^T$. The ideal per-phase back-EMF is trapezoidal, with a 120° plateau and a 60° edge, described by template $T(\theta)$.

$$T(\theta) = \begin{cases} +1, & |\theta| \leq \frac{\pi}{3}, \\ -\frac{6}{\pi}|\theta| + 3, & \frac{\pi}{3} < |\theta| < \frac{2\pi}{3}, \\ -1, & |\theta| \geq \frac{2\pi}{3}. \end{cases} \quad (7)$$

Hence, the ideal phase back EMF is

$$e_k^{\text{ideal}}(\theta_e) = E_m T(\theta_e - \zeta_k), \quad (8)$$

As the conduction interval aligns with the active EMF sector, the torque remains constant under square-wave current:

$$T_e^{\text{ideal}} = \frac{E_m I}{\omega_m}. \quad (9)$$

Figure 2 illustrates this ideal case: the trapezoidal EMFs are aligned with the active conduction sectors, the phase currents remain constant within each sector, and the torque trace is flat over one electrical period.

In practice, BLDC back-EMFs deviate from this ideal form due to slotting effects [16], winding distribution harmonics [17], core saturation [18], air gap eccentricity [19], magnet tolerances [20], and inverter nonlinearities [21]. Each phase can therefore be modeled as

$$e_k(\theta_e) = E_m T(\theta_e - \zeta_k) + \Delta e_k(\theta_e), \quad (10)$$

where $\Delta e_k(\theta_e)$ aggregates all non-ideal contributions.

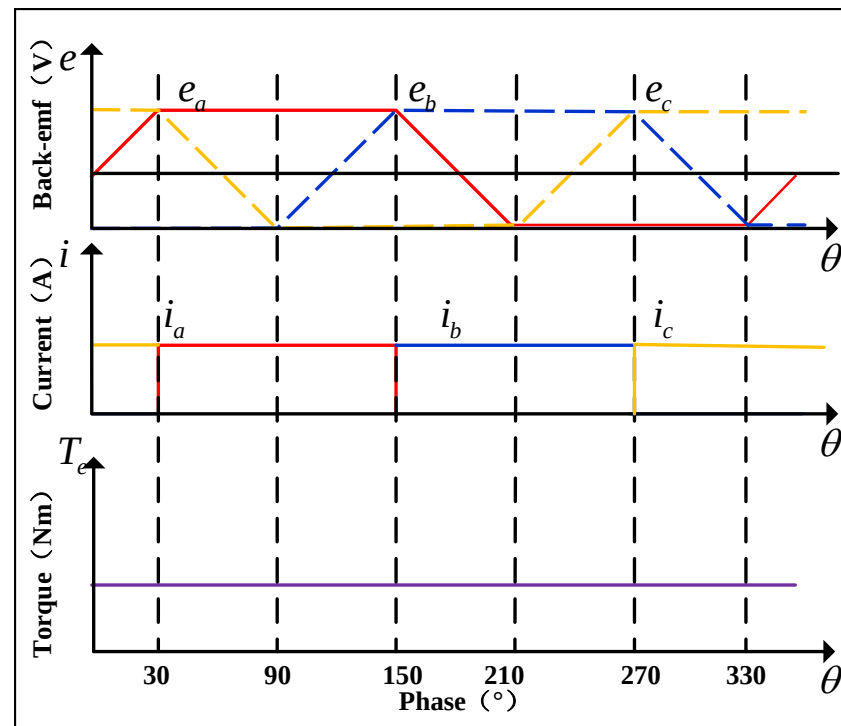


Figure 2. Torque response of ideal square-wave current and back-EMF. The red, blue, and yellow lines represent the back-EMF of phase A, phase B, and phase C, respectively.

This provides a parsimonious parameterization of non-ideal back-EMF. Figure 3 depicts a representative non-ideal back-EMF. Compared with the ideal template, the waveform shows noticeable deviations in both amplitude and shape.

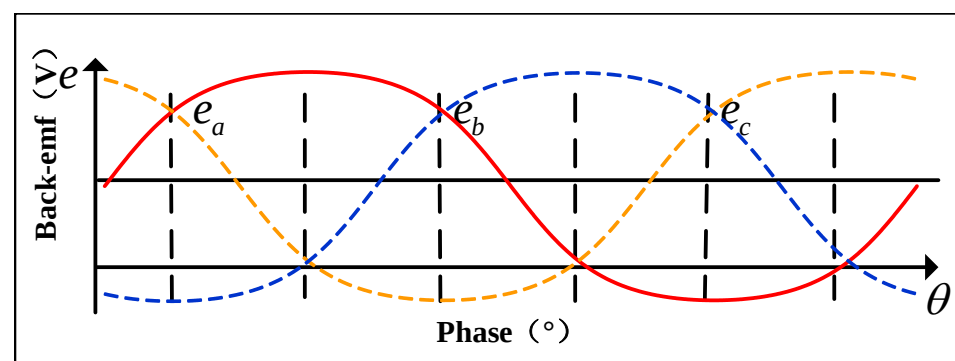


Figure 3. Non-ideal per-phase back-EMF. The red, blue, and yellow lines represent the back-EMF of phase A, phase B, and phase C, respectively.

Substituting (10) into the torque expression under constant-current conduction gives

$$T_e(\theta_e) = \underbrace{\frac{E_m I}{\omega_m}}_{\text{constant baseline}} + \underbrace{\frac{I}{\omega_m} \sum_{k=a,b,c} \Delta e_k(\theta_e) S(\theta_e - \zeta_k)}_{\text{ripple term } T_{\text{rip}}(\theta_e)}. \quad (11)$$

When the same square-wave current is applied, the torque is no longer constant. Figure 4 shows the ripple that emerges within each conduction interval.

Based on the above analysis, precise torque control requires taking the electromagnetic torque as the direct control objective. Accurate torque reconstruction depends on acquiring the instantaneous back-EMF waveform of each phase during every conduction interval. Although offline calibration can provide back-EMF mapping, achieving the torque accuracy required for reaction wheel output necessitates a model that incorporates the parameters that are dynamically affected by operating conditions, such as magnet temperature, microscopic eccentricity, and local magnetic saturation. These parameters are often unmeasurable in practical environments and are highly sensitive to speed, load, ambient temperature, and long-term aging. Due to the above uncertainties and measurement noise, a static offline model gradually loses accuracy as operating conditions change [22]. In response, this article proposes a high-speed, robust, online back-EMF observation method capable of real-time estimation of each phase's e_k [23], enabling precise reconstruction of electromagnetic torque under varying conditions and providing a stable, reliable basis for advanced composite control strategies.

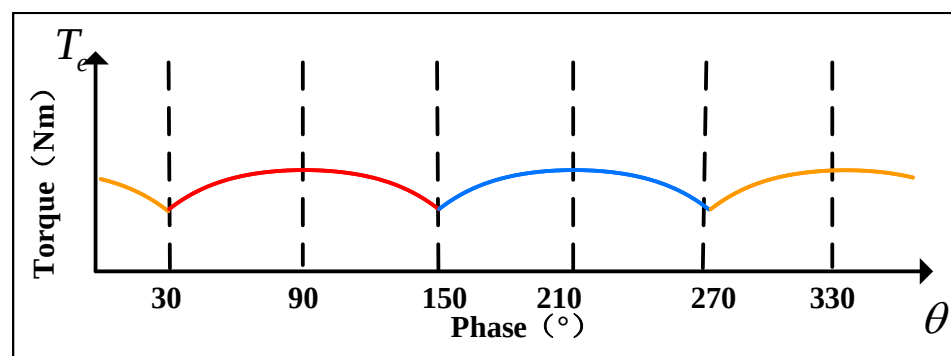


Figure 4. Torque response of ideal square-wave current and non-ideal back-EMF.

3. Design of the Composite Control Strategy

Figure 5 illustrates the proposed composite control architecture for the BLDC reaction wheel drive. The ANFTSMO estimates the EMF, reconstructing the electromagnetic torque and providing high-precision torque feedback. Based on this estimation, the outer PI controller generates the reference current command for the inner loop, which is then precisely regulated by the ASMC to ensure the accurate tracking of the commanded current. The ASMC effectively suppresses chattering through adaptive switching gain. As a result, the output torque can accurately follow the reference torque while significantly reducing the torque ripple caused by non-ideal back-EMF during the conduction interval. In this control architecture, torque serves as both the control objective and the feedback variable, forming a closed-loop system with coordinated inner and outer loops. The system's closed-loop stability is guaranteed through Lyapunov analysis, ensuring stable operation and fast response in various dynamic environments. The detailed design of the observer and controller is discussed in the following subsections.

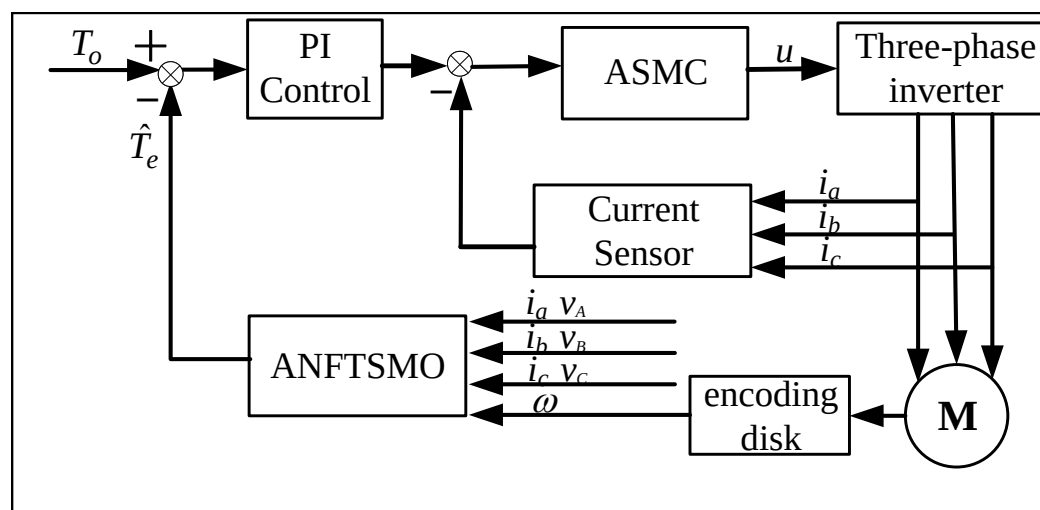


Figure 5. Overall control scheme of the motor drive system.

3.1. Adaptive Nonsingular Fast Terminal Sliding-Mode Observer Design

The proposed ANFTSMO is designed to accurately estimate the phase back-EMF of the BLDC motor under distorted waveform conditions and reconstruct the electromagnetic torque in real time. Compared with traditional SMOs, this design eliminates singularities, reduces chattering, and adapts the observer gain online to accommodate parameter variations. The ANFTSMO can be expressed as follows:

$$\frac{d}{dt} \begin{bmatrix} \hat{i}_a \\ \hat{i}_b \\ \hat{i}_c \end{bmatrix} = -\frac{R}{L} \begin{bmatrix} \hat{i}_a \\ \hat{i}_b \\ \hat{i}_c \end{bmatrix} + \frac{1}{L} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} - \begin{bmatrix} H_a \\ H_b \\ H_c \end{bmatrix}. \quad (12)$$

where $\hat{i}_a, \hat{i}_b, \hat{i}_c$ denote the estimated phase currents; u_a, u_b, u_c represent the corresponding phase terminal voltages; and H_a, H_b, H_c denote the reaching laws of the nonsingular fast terminal sliding-mode observer. To improve convergence and accuracy under non-ideal back-EMF, the observer introduces a nonsingular fast terminal sliding surface:

$$\mathbf{s}_0 = \mathbf{x}_1 + \int j\mathbf{x}_1 dt + \int k\mathbf{x}_1^{\frac{p}{q}} dt, \quad (13)$$

$$\mathbf{x}_1 = \begin{bmatrix} \tilde{i}_a \\ \tilde{i}_b \\ \tilde{i}_c \end{bmatrix} = \begin{bmatrix} \hat{i}_a - i_a \\ \hat{i}_b - i_b \\ \hat{i}_c - i_c \end{bmatrix}. \quad (14)$$

where p and q are positive odd integers, and $p < q$.

Because the integral terms introduce no singularity, j and k are chosen as positive constants. Here, $\mathbf{x}_1$ denotes the current error. The time derivative of the sliding surface can be written as

$$\dot{\mathbf{s}}_0 = \mathbf{x}_2 + j\mathbf{x}_1 + k\mathbf{x}_1^{\frac{p}{q}}. \quad (15)$$

$$\mathbf{x}_2 = \dot{\mathbf{x}}_1 = \begin{bmatrix} \dot{\tilde{i}}_a \\ \dot{\tilde{i}}_b \\ \dot{\tilde{i}}_c \end{bmatrix} = -\frac{R}{L} \begin{bmatrix} \tilde{i}_a \\ \tilde{i}_b \\ \tilde{i}_c \end{bmatrix} + \frac{1}{L} \begin{bmatrix} E_a \\ E_b \\ E_c \end{bmatrix} - \begin{bmatrix} H_a \\ H_b \\ H_c \end{bmatrix}. \quad (16)$$

where $\mathbf{x}_2$ is the derivative of the current error, which represents the rate of change in the current error.

When $\dot{\mathbf{s}}_o = 0$, the variation rate of the system error can be expressed as follows:

$$\dot{\mathbf{x}}_2 = -j\mathbf{x}_1 - k\mathbf{x}_1^{\frac{p}{q}}. \quad (17)$$

By combining (15) and (16), $\dot{\mathbf{s}}_o$ can be obtained as follows:

$$\dot{\mathbf{s}}_o = -\frac{R}{L}\mathbf{x}_1 + \frac{\mathbf{E}}{L} - \mathbf{H} + j\mathbf{x}_1 + k\mathbf{x}_1^{\frac{p}{q}}. \quad (18)$$

where $\mathbf{E} = [E_a \ E_b \ E_c]^T$ is the back-EMF vector, and $\mathbf{H} = [H_a \ H_b \ H_c]^T$ represents the reaching law vector.

To ensure fast convergence and chattering suppression, an adaptive reaching law is adopted:

$$\dot{\mathbf{s}}_o = -k_{1o}\mathbf{s}_o - f_o(\mathbf{s}_o) \operatorname{sgn}(\mathbf{s}_o). \quad (19)$$

The adaptive function $f_o(\mathbf{s}_o)$ dynamically adjusts its amplitude according to the distance of the system state from the sliding surface:

$$f_o(\mathbf{s}_o) = m_o |\mathbf{s}_o| \cdot \frac{1 + \delta_o}{\delta_o + e^{-\varepsilon_o |\mathbf{s}_o|}}. \quad (20)$$

where $m_o > 0$, $0 < \delta_o < 1$, and $\varepsilon_o > 0$. When the system trajectory is far from the sliding surface, $|\mathbf{s}_o|$ increases, and $f_o(\mathbf{s}_o) > m_o$, which accelerates the convergence toward the sliding surface. Conversely, as the trajectory approaches the sliding surface, $f_o(\mathbf{s}_o)$ decreases, effectively suppressing chattering. Exactly on the sliding surface, $f_o(\mathbf{s}_o) = m_o$.

Substituting (18) into (19), the reaching law vector can be expressed as (21)

$$\mathbf{H} = -\frac{R}{L}\mathbf{x}_1 + j\mathbf{x}_1 + k\mathbf{x}_1^{\frac{p}{q}} + k_{1o}\mathbf{s}_o + f_o(\mathbf{s}_o) \operatorname{sgn}(\mathbf{s}_o). \quad (21)$$

Within the structure of the proposed observer, H inherently represents the back-EMF of the corresponding phase, denoted by $\hat{\mathbf{e}}$. This formulation yields high-precision back-EMF estimation even under distorted waveform conditions. The estimated phase EMFs are then utilized to reconstruct the electromagnetic torque as

$$\hat{T}_e = \frac{\hat{e}_a i_a + \hat{e}_b i_b + \hat{e}_c i_c}{\omega_m}. \quad (22)$$

where $\hat{e}_a$, $\hat{e}_b$, and $\hat{e}_c$ are the estimated phase EMFs.

The stability of the ANFTSMO is analyzed using Lyapunov theory.

Choosing the Lyapunov function as

$$V = \frac{1}{2} \mathbf{s}_o^T \mathbf{s}_o. \quad (23)$$

Its time derivative is given by

$$\dot{V} = \mathbf{s}_o^T \dot{\mathbf{s}}_o = -k_{1o} \mathbf{s}_o^T \mathbf{s}_o - m_o |\mathbf{s}_o| \left(\frac{1 + \delta_o}{\delta_o + e^{-\varepsilon_o |\mathbf{s}_o|}} \right). \quad (24)$$

Since $k_{1o} > 0$, $0 < \delta_o < 1$, and $m_o > 0$, it follows from (24) that $\dot{V} \leq 0$. Hence, the proposed observer satisfies the Lyapunov stability condition. Moreover, because $\dot{V} = 0$ only when $\mathbf{s}_o = 0$, by LaSalle's invariance principle, the trajectories converge to the largest invariant set contained in $\{\dot{V} = 0\}$, i.e., $\mathbf{s}_o = 0$. As the nonsingular sliding surface is designed such that $\mathbf{s}_o = 0 \iff \mathbf{e} = 0$, the estimation error $\mathbf{e}$ tends to zero. Therefore, the proposed ANFTSMO is asymptotically stable, and the estimated back-EMFs

and reconstructed torque converge to their true values. This ensures that the estimated quantities are bounded and continuous in real time, providing reliable feedback for the inner current controller. The overall framework of the ANFTSMO is illustrated in Figure 6.

The proposed ANFTSMO effectively combines nonsingular terminal sliding mode theory with an adaptive gain adjustment mechanism. By dynamically regulating the switching gain according to the distance between the system states and the sliding surface, the observer achieves fast convergence when the estimation error is large and suppresses chattering near the steady state region. Lyapunov stability ensures that the estimated quantities are bounded and continuous, providing reliable torque feedback for subsequent current and rate control loops. This design thus enables high-precision back-EMF sensing even under significantly distorted waveforms.

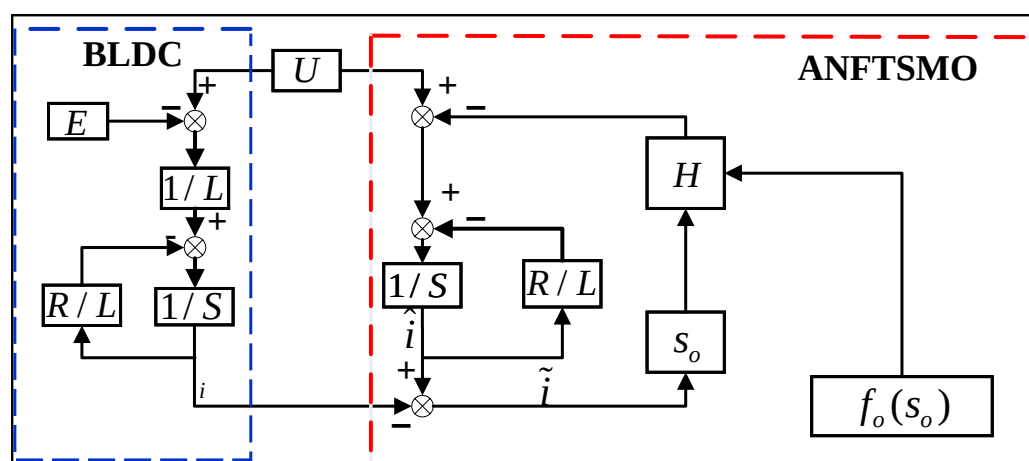


Figure 6. Control block diagram of ANFTSMO. The blue curved box represents the BLDC system, and the red dashed box represents the observer system.

3.2. Adaptive Sliding-Mode Current Control Design

The inner current-loop controller is designed to regulate the phase current of the BLDC reaction wheel motor in real time. Under the conventional three-phase half-bridge commutation scheme, only one stator phase conducts current during each 60° electrical interval. The dynamic equation of the conducting phase $k \in \{a, b, c\}$ is written as

$$L \frac{di_k}{dt} = u_k - Ri_k + E_k. \quad (25)$$

The control objective is to ensure accurate current tracking:

$$i_k \rightarrow i_k^*. \quad (26)$$

where i_k^* is the reference current provided by the outer torque-rate feedback loop. Define the current-tracking error vector as

$$\mathbf{e} = \mathbf{i} - \mathbf{i}^* = \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} = \begin{bmatrix} i_a - i_a^* \\ i_b - i_b^* \\ i_c - i_c^* \end{bmatrix}. \quad (27)$$

An integral sliding surface is introduced to remove steady-state bias:

$$\mathbf{s}_c = \mathbf{e} + \lambda \int \mathbf{e} dt. \quad (28)$$

where $\lambda > 0$ determines the current-loop bandwidth and convergence rate.

To guarantee fast convergence and suppress chattering, an adaptive exponential reaching law is introduced:

$$\dot{\mathbf{s}}_c = -\alpha_c \mathbf{s}_c - f_c(\mathbf{s}_c) \operatorname{sgn}(\mathbf{s}_c). \quad (29)$$

where $\alpha_c > 0$ is the linear convergence gain.

The adaptive switching gain function $f_c(\mathbf{s}_c)$ dynamically regulates its amplitude according to the distance from the sliding surface:

$$f_c(\mathbf{s}_c) = m_c |\mathbf{s}_c| \cdot \frac{1 + \delta_c}{\delta_c + e^{-\varepsilon_c |\mathbf{s}_c|}}, \quad 0 < \delta_c < 1, \varepsilon_c > 0, m_c > 0. \quad (30)$$

When $|\mathbf{s}_c|$ is large, $f_c(\mathbf{s}_c)$ increases to accelerate convergence; when $|\mathbf{s}_c|$ is small, $f_c(\mathbf{s}_c)$ decreases, thereby suppressing chattering near the sliding surface. Exactly on the sliding surface, $f_c(\mathbf{s}_c) = m_c |\mathbf{s}_c|$. Differentiating (28) yields

$$\dot{\mathbf{s}}_c = \dot{\mathbf{e}} + \lambda \mathbf{e} = \dot{\mathbf{i}} - \dot{\mathbf{i}}^* + \lambda \mathbf{e}. \quad (31)$$

Substituting the current dynamic model (25) gives

$$\dot{\mathbf{s}}_c = \frac{1}{L} (u_k - Ri_k + E_k) - \dot{\mathbf{i}}^* + \lambda \mathbf{e}. \quad (32)$$

By enforcing (29), the control input can be obtained as

$$u_k = Ri_k + E_k + L_{\text{eq}} (\dot{\mathbf{i}}^* - \lambda \mathbf{e} - \alpha_c \mathbf{s}_c - f_c(\mathbf{s}_c) \operatorname{sgn}(\mathbf{s}_c)). \quad (33)$$

Since the actual back-EMF is not measurable in practice, the estimated value obtained from the ANFTSMO is used as a feed-forward compensation. The implementable control law is

$$u_k = Ri_k + \hat{E}_k + L_{\text{eq}} (\dot{\mathbf{i}}^* - \lambda \mathbf{e} - \alpha_c \mathbf{s}_c - f_c(\mathbf{s}_c) \operatorname{sgn}(\mathbf{s}_c)). \quad (34)$$

The stability of the proposed ASMC is analyzed using Lyapunov theory. Consider the Lyapunov function as

$$V_c = \frac{1}{2} \mathbf{s}_c^T \mathbf{s}_c. \quad (35)$$

Its time derivative is given by

$$\dot{V}_c = \mathbf{s}_c^T \dot{\mathbf{s}}_c = -\alpha_c \mathbf{s}_c^T \mathbf{s}_c - |\mathbf{s}_c^T| f_c(\mathbf{s}_c) \operatorname{sgn}(\mathbf{s}_c). \quad (36)$$

Since $\alpha_c > 0$, $f_c(\mathbf{s}_c) > 0$, and $0 < \delta_c < 1$, it follows that $\dot{V}_c \leq 0$. Equality holds only when $\mathbf{s}_c = 0$. Therefore, by LaSalle's invariance principle, $\mathbf{e} \rightarrow 0$, and the current-tracking error converges asymptotically.

The ASMC thus ensures the robust asymptotic convergence of the current loop despite parameter uncertainties and EMF distortion. The coupling between the ASMC and ANFTSMO subsystems is discussed next to demonstrate the global asymptotic stability of the integrated system.

The closed-loop dynamics can be regarded as an interconnected system in which the current control subsystem is driven by the back-EMF estimation error of the observer. Substituting the control law (33) into the plant model yields

$$L \dot{i}_k = L (\dot{i}_k^* - \lambda \mathbf{e}_i - \alpha_c \mathbf{s}_c - f_c(\mathbf{s}_c) \operatorname{sgn}(\mathbf{s}_c)) + (\hat{E}_k - E_k). \quad (37)$$

where the last term represents the estimation error $\tilde{E}_k = \hat{E}_k - E_k$ introduced by the observer.

The estimation error dynamics governed by the ANFTSMO have already been proven to be asymptotically stable, ensuring that $\tilde{E}_k$ is bounded and converges to zero.

To analyze the stability of the coupled system, consider a composite Lyapunov function:

$$V = V_c + \gamma V_o = \frac{1}{2} \mathbf{s}_c^T \mathbf{s}_c + \frac{\gamma}{2} \mathbf{s}_o^T \mathbf{s}_o, \quad \gamma > 0. \quad (38)$$

where V_c and V_o correspond to the Lyapunov functions of the current controller and the observer, respectively.

Differentiating along the closed-loop trajectories yields

$$\begin{aligned} \dot{V} \leq & -(\alpha_c - \mu) \mathbf{s}_c^T \mathbf{s}_c - \gamma k_{1o} \mathbf{s}_o^T \mathbf{s}_o - |\mathbf{s}_c| f_c(\mathbf{s}_c) \operatorname{sgn}(\mathbf{s}_o) \\ & + \frac{\mu}{2L^2} \tilde{E}_k^2. \end{aligned} \quad (39)$$

where $\mu > 0$ is a small constant.

As $\tilde{E}_k \rightarrow 0$ according to the observer dynamics, there exists a positive constant $\eta > 0$ such that

$$\dot{V} \leq -\eta (\|\mathbf{s}_c\|^2 + \|\mathbf{s}_o\|^2). \quad (40)$$

Consequently, both $\mathbf{s}_c$ and $\mathbf{s}_o$ converge asymptotically to zero, implying that the current-tracking error and the back-EMF estimation error simultaneously vanish. Hence, the proposed ANFTSMO–ASMC framework is asymptotically stable as an integrated system and ensures precise torque regulation under non-ideal back-EMF conditions and parameter uncertainties.

4. Simulation and Experimental Results

4.1. Experimental Platform and Parameters

The reaction wheel testbed consisted of a 24 V DC bus, a three-phase half-bridge inverter, and a BLDC reaction wheel motor. The experimental platform is shown in Figure 7. A TMS320F280049C DSP executed the proposed ANFTSMO–ASMC scheme at 20 kHz. Phase currents and DC bus voltage were measured using current and voltage sensors, and rotor position and speed were obtained from an encoder. The plant and controller parameters used in both simulation and experiment are listed in Tables 1 and 2, respectively.

Table 1. Parameters of the BLDC reaction wheel system.

Parameter	Value	Parameter	Value
Rated speed (r/min)	6000	Armature inductance L (μH)	100
DC bus voltage V_{dc} (V)	24	Number of pole pairs	8
Rotor inertia J ($\text{kg}\cdot\text{m}^2$)	0.00956	Back-EMF constant ($\text{V}/(\text{r}/\text{min})$)	0.00435
Armature resistance R (Ω)	0.942	Maximum current $I_{\max}$ (A)	2.5

Table 2. Parameters of the proposed controller.

Parameter	Value	Parameter	Value
j	10	ε_o	10^{-3}
k	5	λ	1000
p	1	α_c	2
q	2	m_c	50
k_{1o}	5	δ_c	0.1
m_o	5	ε_c	10^{-3}
δ_o	0.1		

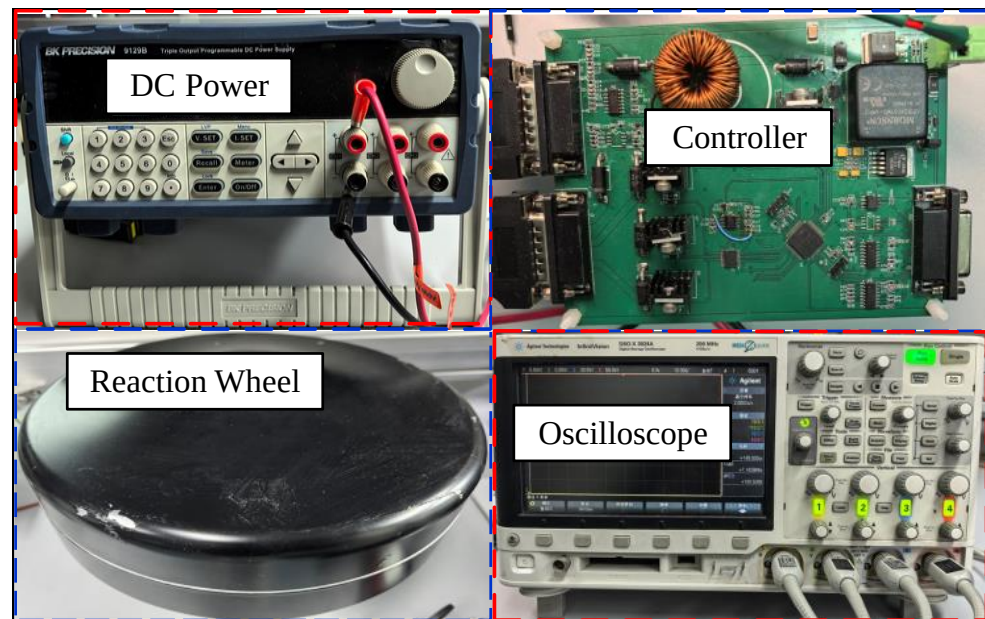


Figure 7. Experimental platform of the reaction wheel system.

4.2. Simulation and Experimental Validations

In the simulation, the phase back-EMF estimation performance of the ANFTSMO and the conventional SMO was compared under distorted back-EMF conditions, as shown in Figure 8. The results indicate that the ANFTSMO achieves faster and more accurate tracking of the actual waveform while effectively suppressing high-frequency chattering. These findings confirm its advantages in dynamic response and robustness, providing a reliable foundation for high-precision torque estimation.

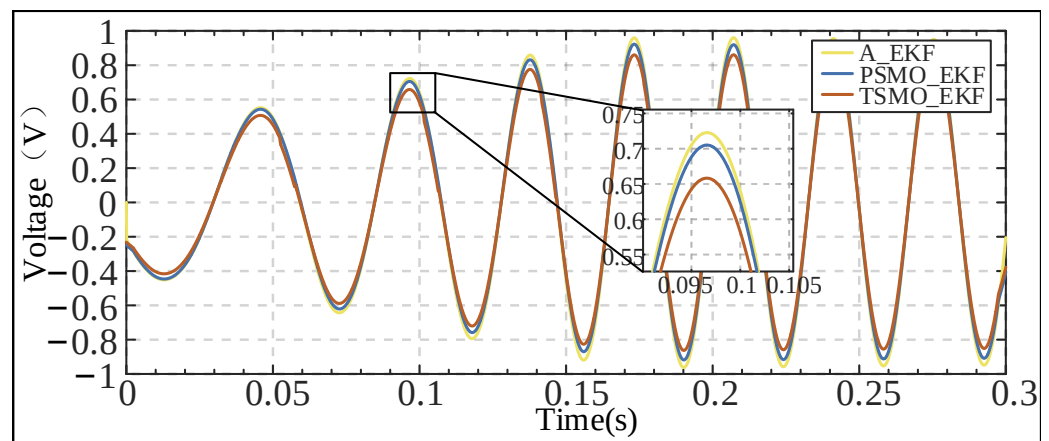


Figure 8. Back-EMF estimation performance of SMO and ANFTSMO.

As shown in Figure 9, the simulated results of the phase current and electromagnetic torque responses under the three control strategies were compared. The left column corresponds to the conventional constant current PI control, the middle column represents the ANFTSMO-based torque control with PI current regulation, and the right column illustrates the proposed ANFTSMO–ASM composite control.

Under conventional PI control, the phase current remains relatively smooth; however, the non-ideal back-EMF waveform causes pronounced periodic torque oscillations with a peak-to-peak amplitude of approximately 0.03 N·m. When the ANFTSMO is introduced, the electromagnetic torque can be reconstructed and used as a feedback signal in the control loop in which the current controller is implemented as a PI regulator. This torque

based feedback effectively aligns the current waveform with the distorted back-EMF, thereby mitigating torque ripple within the conduction intervals. Consequently, the torque waveform becomes smoother, and the steady-state tracking error decreases to around 3%. After replacing the PI regulator with the ASMC, the current response exhibits an even smoother profile. The torque closely follows the reference across the entire conduction period, the ripple amplitude is further suppressed, and the steady-state error remains below 1%. These results verify the theoretical feasibility and superior dynamic performance of the proposed ANFTSMO–ASMC composite control.

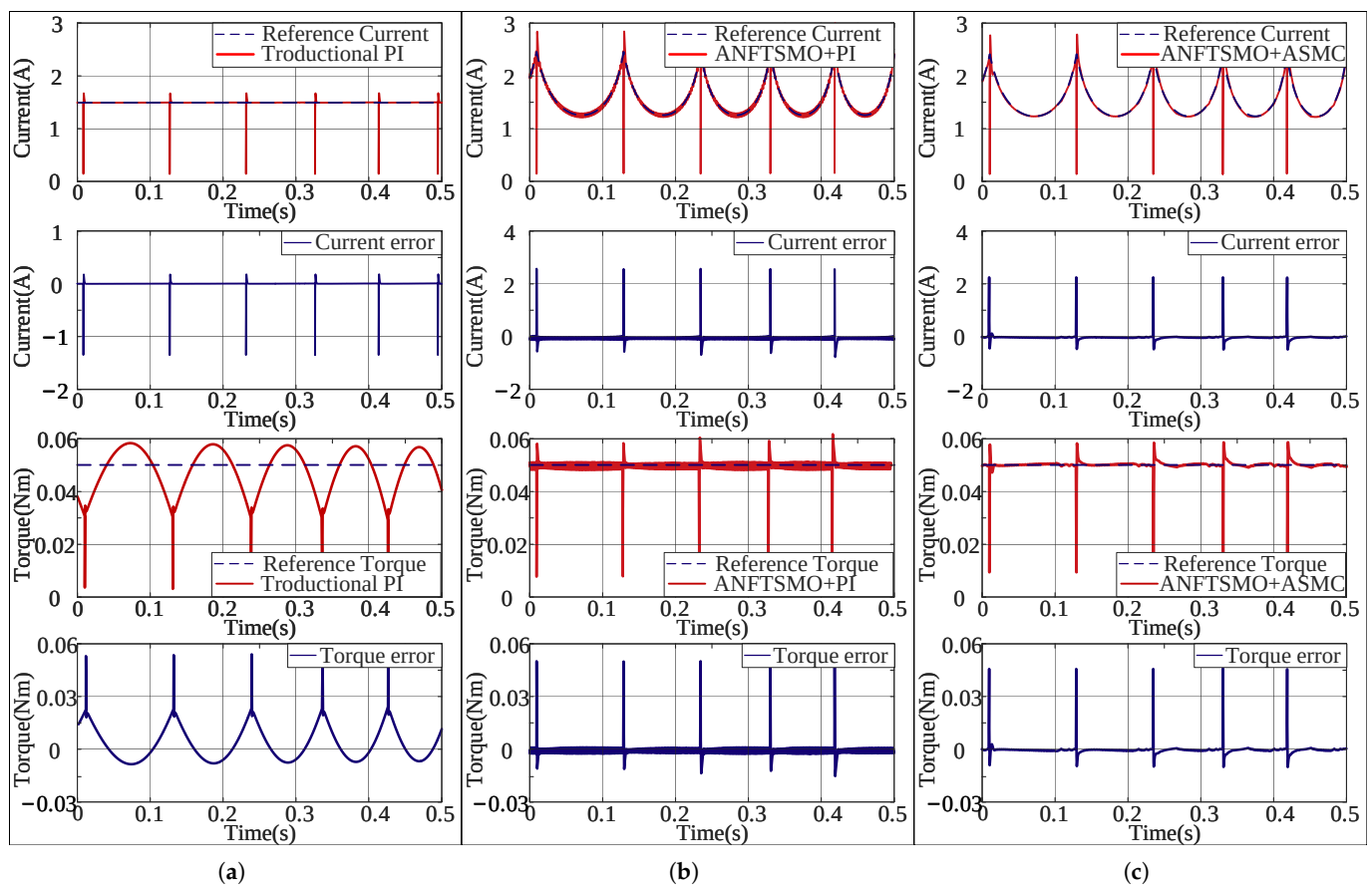


Figure 9. Simulated phase current and torque waveforms under (a) conventional current PI, (b) ANFTSMO + PI, and (c) ANFTSMO + ASMC control.

To further confirm the practical effectiveness of the proposed control strategy, the same three control schemes were implemented on an experimental reaction wheel platform, as shown in Figure 7, with the measured waveforms presented in Figure 10 and the specific experimental data listed in Table 3. The experimental trends are consistent with the simulation results: the constant-current PI method produces evident commutation-related ripple, whereas the ANFTSMO-based PI control significantly improves torque smoothness and reduces the steady-state tracking error to about 6%. When the ANFTSMO is combined with ASMC current regulation, the current waveform becomes smoother, and the torque tracks the reference almost perfectly throughout all conduction intervals. The conduction interval ripple magnitude is more than 65% lower than that of the PI method, and the steady-state error remains within 2.7%. These experimental findings validate that the proposed ANFTSMO–ASMC composite control achieves precise torque regulation, provides strong robustness, and produces stable real-time performance under distorted back-EMF and mechanical disturbance conditions.

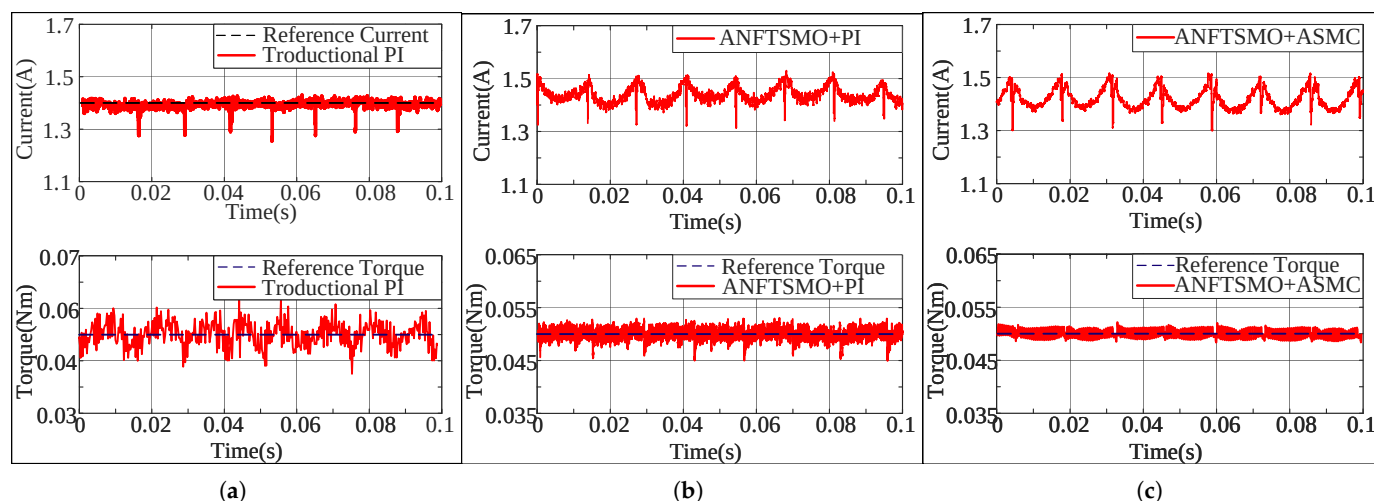


Figure 10. Experimental phase current and torque waveforms under (a) constant-current PI, (b) ANFTSMO + PI, and (c) ANFTSMO + ASMC control.

Table 3. Experimental comparison of torque ripple suppression performance. Note: The bold values indicate the best performance.

Control Strategy	Peak-to-Peak Ripple (N·m)	Steady-State Error (%)	Reduction (%)
Traditional Constant Current PI	0.020	13.3	-
ANFTSMO + PI	0.009	6.0	55.0
ANFTSMO + ASMC	0.004	2.7	80.0

5. Conclusions

This article presents a composite torque control strategy for reaction wheel systems that integrates an ANFTSMO, an ASMC, and an outer rate feedback loop. The proposed framework shifts the control paradigm from conventional current regulation to direct torque control, effectively addressing conduction interval torque ripple under non-ideal back-EMF conditions. The ANFTSMO ensures accurate back-EMF estimation and high-fidelity electromagnetic torque reconstruction; the ASMC provides fast and robust current regulation with adaptive gain adjustment to suppress chattering and to handle nonlinear dynamics; and the rate feedback loop compensates for frictional- and load-induced disturbances, ensuring that the actual torque remains tightly aligned with its command. Simulation and experimental results verify the superiority of the proposed method, demonstrating enhanced dynamic performance, significant torque ripple reduction, and improved disturbance rejection.

Future work will focus on further optimization of the adaptive gain mechanism and the extension of this approach to complex aerospace applications characterized by strong nonlinearities and external disturbances.

Author Contributions: Conceptualization, Z.W. and H.L. (Haitao Li 1); methodology, Z.W.; software, Z.W.; validation, Z.W.; investigation, Z.W.; writing—original draft preparation, Z.W.; visualization, Z.W. and H.L. (Haitao Li 2); writing—review and editing, T.W. and X.C.; supervision, T.W.; funding acquisition, T.W. and H.L. (Haitao Li 1). All authors have read and agreed to the published version of this manuscript.

Funding: This research was funded by the National Natural Science Foundation of China (grant numbers 52301071, 62573027, and 62173015) and the “Key Innovation Yongjiang 2035” project. The APC was funded by the National Natural Science Foundation of China.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within this article.

Acknowledgments: The authors would like to thank the anonymous reviewers for their valuable comments and suggestions, which have greatly improved the quality of this paper.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ANFTSMO	Adaptive Nonsingular Fast Terminal Sliding-Mode Observer
ASMC	Adaptive Sliding-Mode Controller
Back-EMF	Back-Electromotive Force
BLDC	Brushless DC
DTC	Direct Torque Control
EMI	Electromagnetic Interference
PMSM	Permanent Magnet Synchronous Motor
PWM	Pulse-Width Modulation

References

- Sanfedino, F.; Thiébaud, G.; Alazard, D.; Guercio, N.; Deslaef, N. Advances in fine line-of-sight control for large space flexible structures. *Aerosp. Sci. Technol.* **2022**, *130*, 107961. [\[CrossRef\]](#)
- Srinivasan, J.M.; Barltrop, C.; Berman, S.; Bushman, S.; Dickson, J.; Drain, T.; Engelbrecht, C.; Fugett, D.; Gruel, D.; Hill, S.; et al. Europa Clipper flight system overview. *Space Sci. Rev.* **2025**, *221*, 14. [\[CrossRef\]](#)
- Zhang, Z.; Li, Y. An AEFA-based optimum design of fuzzy PID controller for attitude control flywheel with BLDC motor. *Aerospace* **2022**, *9*, 789. [\[CrossRef\]](#)
- Salmanfar, M.; Alizadeh Pahlavani, M.R.; Dehestani Kolagar, A.; KoohMaskan, Y. Position control of a permanent magnet brushless DC motor using a model predictive control method with Laguerre functions. *J. Electr. Syst.* **2024**, *20*, 3551–3564.
- Zeng, C.; Wan, Z.; Huang, S.; He, J. Torque ripple suppression in non-commutation interval of the coreless brushless DC motor based on unipolar PWM predictive control. *IET Power Electron.* **2024**, *17*, 1621–1631. [\[CrossRef\]](#)
- Han, Q.; Gao, S.; Chu, F. Micro-vibration analysis, suppression, and isolation of spacecraft flywheel rotor systems: A review. *Vibration* **2024**, *7*, 229–263. [\[CrossRef\]](#)
- Mahmouditabar, F.; Vahedi, A.; Takorabet, N. Robust Design of BLDC Motor Considering Driving Cycle. *IEEE Trans. Transp. Electr.* **2023**, *10*, 1414–1424. [\[CrossRef\]](#)
- Zhang, H.; Dou, M.; Yan, L. A novel current solution for torque ripple suppression of BLDCM with minimal rotor eddy-current loss. *IEEE Trans. Power Electron.* **2020**, *35*, 2831–2842. [\[CrossRef\]](#)
- He, J.; Yan, C.; Wang, X. Torque ripple suppression of brushless DC motor drive system based on improved harmonic injection active disturbance rejection control. *Sensors* **2022**, *22*, 1069. [\[CrossRef\]](#)
- Son, D.I.; Han, J.S.; Park, J.S.; Lim, H.S.; Lee, G.H. Performance improvement of DTC-SVM of PMSM with compensation for the dead time effect and power switch loss based on extended Kalman filter. *Electronics* **2023**, *12*, 966. [\[CrossRef\]](#)
- Sa, X.; Wu, H.; Zhao, G.; Zhao, Z. Torque ripple reduction in BLDC motors using phase current integration and enhanced zero vector DTC. *Electronics* **2025**, *14*, 2999. [\[CrossRef\]](#)
- Xu, W.; Qu, S.; Zhao, L.; Zhang, H. An improved adaptive sliding mode observer for middle- and high-speed rotor tracking. *IEEE Trans. Power Electron.* **2021**, *36*, 1043–1053. [\[CrossRef\]](#)
- Wang, Y.; Feng, Y.; Zhang, X.; Liang, J. A new reaching law for antidisturbance sliding-mode control of PMSM speed regulation system. *IEEE Trans. Power Electron.* **2020**, *35*, 4117–4126. [\[CrossRef\]](#)
- Palanisamy, S.; Chenniappan, S.; Nayagi, A.T.; Mahendran, R. Introduction to electric power systems. In *Basic Electrical and Instrumentation Engineering*; Wiley: Hoboken, NJ, USA, 2021; pp. 1–77.
- Nayak, D.S.; Shivarudraswamy, R. Loss and efficiency analysis of BLDC motor and universal motor by mathematical modelling in the mixer grinder. *J. Inst. Eng. India Ser. B* **2022**, *103*, 517–523. [\[CrossRef\]](#)
- Jarso, G.C.; Nallamotheu, R.B.; Gopal, R.; Jin, G.G. Strategic slot-pole optimization in electromagnetic coupling with random forest regressor model inspired by response surface methodology. *Results Eng.* **2025**, *27*, 105640. [\[CrossRef\]](#)
- Huang, P.-S.; Tsai, C.-T.; Hwang, J.-C.; Lin, C.-T.; Lin, Y.-T. Current harmonics suppression of six-phase permanent-magnet synchronous motor drives using back-electromotive-force harmonics compensation. *Energies* **2024**, *17*, 6280. [\[CrossRef\]](#)

18. Zhang, H.; Fan, M.; Qiao, J.; He, X.; Yang, M.; Tuo, J. Analytical prediction of electromagnetic performance for surface-embedded permanent magnet in-wheel machines considering iron's nonlinearity. *Sci. Rep.* **2024**, *14*, 26353. [[CrossRef](#)] [[PubMed](#)]
19. Feng, L.; Yu, S.; Zhang, F.; Jin, S.; Sun, Y. Study on performance of low-speed high-torque permanent magnet synchronous motor with dynamic eccentricity rotor. *Energy Rep.* **2022**, *8*, 1421–1428. [[CrossRef](#)]
20. Popsi, N.R.S.; Anik, A.; Verma, R.; Viana, C.; Iyer, K.L.V.; Kar, N.C. Influence of electric motor manufacturing tolerances on end-of-line testing: A review. *Energies* **2024**, *17*, 1913. [[CrossRef](#)]
21. Kivanc, O.C.; Ozturk, S.B.; Toliyat, H.A. On-line dead time compensator for PMSM drive based on current observer. *Eng. Sci. Technol. Int. J.* **2022**, *25*, 100987. [[CrossRef](#)]
22. Zhu, Z.; Liang, D.; Liu, K. Online Parameter Estimation for Permanent Magnet Synchronous Machines: An Overview. *IEEE Access* **2021**, *9*, 59059–59084. [[CrossRef](#)]
23. Zeghlache, A.; Djerioui, A.; Mekki, H.; Zeghlache, S.; Benkhoris, M.F. Robust Sensorless PMSM Control with Improved Back-EMF Observer and Adaptive Parameter Estimation. *Electronics* **2025**, *14*, 1238. [[CrossRef](#)]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.